being preferred for long time-delay processes, and it confirms that GPC is simple to implement and to use.

The GPC method is a further generalization of the well-known GMV approach, and so can be equipped with design polynomials and transfer-functions which have interpretations as in the GMV case. The companion paper Part II, which follows, explores these ideas and presents simulations which show how GPC can be used for more demanding control tasks.

## REFERENCES


Åström, K. J. and B. Wittenmark (1973). On self-tuning regulators. *Automatica*, **9**, 185–199.

Åström, K. J. and B. Wittenmark (1980). Self-tuning controllers based on pole-zero placement. *Proc. IEE*, **127D**, 120–130.

Bertsekas, D. P. (1976). *Dynamic Programming and Stochatic Control.* Academic Press, New York.

Clarke, D. W. (1984). Self-tuning control of nonminimum-phase systems. *Automatica*, **20**, 501–517.

Clarke, D. W. and P. J. Gawthrop (1975). Self-tuning controller. *Proc. IEE*, **122**, 929–934.

Clarke, D. W. and P. J. Gawthrop (1979). Self-tuning control. *Proc. IEE*, **126**, 633–640.

Clarke, D. W. and L. Zhang (1985). Does long-range predictive control work? IEE Conference "Control 85", Cambridge.

Cutler, C. R. and B. L. Ramaker (1980). *Dynamic Matrix Control—A Computer Control Algorithm.* JACC, San Francisco.

De Keyser, R. M. C. and A. R. Van Cauwenberghe (1982). Typical application possibilities for self-tuning predictive control. *IFAC Symp. Ident. Syst. Param. Est.*, Washington.

De Keyser, R. M. C. and A. R. Van Cauwenberghe (1983). Microcomputer-controlled servo system based on self-adaptive long-range prediction. Advances in Measurement and Control—MECO 1983.

De Keyser, R. M. C. and A. R. Van Cauwenberghe (1985). Extended prediction self-adaptive control. *IFAC Symp. Ident. Syst. Param. Est.*, York.

Kailath, T. (1980). *Linear Systems.* Prentice-Hall, Englewood Cliffs, NJ.

Kurz, H. and W. Goedecke (1981). Digital parameter-adaptive control of processes with unknown dead time. *Automatica*, **17**, 245–252.

Montague, G. A. and A. J. Morris (1985). Application of adaptive control: a heat exchanger system and a penicillin fermentation process. Third workshop on the theory and application of self-tuning and adaptive control, Oxford, U.K.

Mosca, E., G. Zappa and C. Manfredi (1984). Multistep horizon self-tuning controllers: the MUSMAR approach. IFAC 9th World Congress, Budapest, Hungary.

Peterka, V. (1984). Predictor-based self-tuning control. *Automatica*, **20**, 39–50.

Richalet, J., A. Rault, J. L. Testud and J. Papon (1978). Model predictive heuristic control: applications to industrial processes. *Automatica*, **14**, 413–428.

Schweppe, F. C. (1973). *Uncertain Dynamic Systems.* Prentice Hall, Englewood Cliffs, NJ.

Tuffs, P. S. and D. W. Clarke (1985). Self-tuning control of offset: a unified approach. *Proc. IEE*, **132D**, 100–110.

Wellstead, P. E., D. Prager and P. Zanker (1979). Pole assignment self-tuning regulator. *Proc. IEE*, **126**, 781–787.

Ydstie, B. E. (1984). Extended horizon adaptive control. IFAC 9th World Congress, Budapest, Hungary.


## APPENDIX A. MINIMIZATION PROPERTIES OF GPC

Consider the performance index

$$J = (\mathbf{y} - \mathbf{w})^T(\mathbf{y} - \mathbf{w}) + \lambda \tilde{\mathbf{u}}^T\tilde{\mathbf{u}} \quad \text{where } \mathbf{y} = \mathbf{G}\tilde{\mathbf{u}} + \mathbf{f} + \mathbf{e}$$

where $\mathbf{y}$, $\mathbf{w}$, $\tilde{\mathbf{u}}$ are as defined in the main text and

$$\mathbf{e} = [E_1\xi(t+1), E_2\xi(t+2), \ldots, E_N\xi(t+N)]^T.$$

Clearly as $\mathbf{e}$ is a stochastic process the expected value of the cost subject to appropriate conditions must be minimized. The cost to be minimized therefore becomes

$$J_1 = E\{(\mathbf{y} - \mathbf{w})^T(\mathbf{y} - \mathbf{w}) + \lambda \tilde{\mathbf{u}}^T\tilde{\mathbf{u}}\}.$$

Two different assumptions will be made in the next section about the class of admissible controllers for the minimization of the cost above. Only models of autoregressive type are considered:

$$A(q^{-1})y(t) = B(q^{-1})u(t-1) + \xi(t)/\Delta.$$

In the general case of CARIMA models where the disturbances are non-white the predictions $f(t+i)$ are only optimal asymptotically as they have poles at the zeros of the $C(q^{-1})$ and the effect of initial conditions is only eliminated as time tends to infinity. The closed-loop policy requires an optimal Kalman filter with time-varying filter gains, see Schweppe (1973); the parameters of the $C$-polynomial are only the gains of the steady-state Kalman filter once convergence is attained. Thus the closed-loop and the open-loop-feedback policies examined below are different in the case of coloured noise.

### A.1. RELATION OF GPC AND OPEN-LOOP-FEEDBACK-OPTIMAL CONTROL

In this strategy it is assumed that the future control signals are independent of future measurements (i.e. all will be performed in open-loop) and we calculate the set $\tilde{\mathbf{u}}$ such that it minimizes the cost $J_1$. The first control in the sequence is applied and at the next sample the calculations are repeated. The cost becomes:

$$J_1 = E\{\tilde{\mathbf{u}}^T(\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I})\tilde{\mathbf{u}} + 2\tilde{\mathbf{u}}^T(\mathbf{G}^T(\mathbf{f} - \mathbf{w}) + \mathbf{G}^T\mathbf{e}) + \mathbf{f}^T\mathbf{f} + \mathbf{e}^T\mathbf{e} + \mathbf{w}^T\mathbf{w} + 2\mathbf{f}^T(\mathbf{e} - \mathbf{w}) - 2\mathbf{e}^T\mathbf{w}\}.$$

Because of the assumption above, $E\{\tilde{\mathbf{u}}^T\mathbf{G}^T\mathbf{e}\} = 0$; note also that the cost is only affected by the first two terms. By putting the first derivative of the cost equal to zero one obtains:

$$\tilde{\mathbf{u}} = (\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I})^{-1}\mathbf{G}^T(\mathbf{w} - \mathbf{f}).$$

### A.2. RELATION OF GPC AND CLOSED-LOOP-FEEDBACK-OPTIMAL CONTROL

Choose the set $\tilde{\mathbf{u}}$ such that the cost $J_1$ is minimized subject to the condition that $\Delta u(i)$ is only dependent on data up to time $i$. This is the statement of causality. In order to find the set $\tilde{\mathbf{u}}$ assume that by some means $\{\Delta u(t), \Delta u(t+1), \ldots, \Delta u(t+N-2)\}$ is available.

$$\begin{aligned} J_1 = E\{&(\Delta u(t), \ldots, \\ &\Delta u(t+N-1))(\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I})(\Delta u(t), \ldots, \\ &\Delta u(t+N-1))^T \\ &+ 2\tilde{\mathbf{u}}^T(\mathbf{G}^T(\mathbf{f} - \mathbf{w})) \\ &+ 2(\Delta u(t), \ldots, \Delta u(t+N-1))\mathbf{G}^T\mathbf{e}\} \\ &+ \text{extra terms independent of minimization.} \end{aligned}$$

Note that in evaluating $\Delta u(t+N-1)$ the last row $\mathbf{G}^T\mathbf{e}$ is $g_0\sum e_i\xi(t+N-i)$ and $E\{\Delta u(t+N-1)\xi(t+N)\} = 0$ by the assumption of partial information. Therefore $\Delta u(t+N-1)$ can be calculated assuming $\xi(t+N) = 0$. From general state-space considerations it is evident that when minimizing a quadratic cost, assuming a linear plant model with additive noise, the resulting controller is a linear function of the available measurements (i.e. $u(t) = -\mathbf{k}\hat{\mathbf{x}}(t)$ where $\mathbf{k}$ is the feedback gain and $\hat{\mathbf{x}}(t)$ is the state estimate from the appropriate Kalman filter).

$$\begin{aligned} \Delta u(t+N-1) = \text{linear function}\,(&\Delta u(t+N-2), \\ &\Delta u(t+N-3), \ldots, \xi(t+N-1), \\ &\xi(t+N-2), \ldots, y(t), y(t-1), \ldots). \end{aligned}$$

In calculating $\Delta u(t+N-2)$ the following expectations are zero:

$$E\{\Delta u(t+N-2)\xi(t+N)\} = 0,$$
$$E\{\Delta u(t+N-2)\xi(t+N-1)\} = 0.$$

In order to work out $E\{\Delta u(t+N-1)\Delta u(t+N-2)\}$, the part of the expectation associated with $\xi(t+N-1)$ is set to zero as the control signal $\Delta u(t+N-2)$ cannot be a function of $\xi(t+N-1)$. As $\Delta u(t+N-1)$ is a linear function of $\xi(t+N-1)$ this implies that as far as calculation of $\Delta u(t+N-2)$ is concerned, $\Delta u(t+N-1)$ could be calculated by assuming that $\xi(t+N-1)$ and $\xi(t+N)$ were zero. Similarly, the same argument applies for the rest of control signals at each step $i$ assuming $E\{\Delta u(t+i)\xi(t+j)\} = 0$ for $j > i$. This implies that each control signal $\Delta u(t+i)$ is the $i$th control in the sequence assuming that $\xi(t+j) = 0$ for $j > i$. Hence for $\Delta u(t)$ the solution amounts to calculating the first control signal setting $\xi(t+j)$ for $j > 0$ to zero—the same as that considered in part I for an Open-Loop-Feedback-Optimal controller. Note that in the case of $NU < N_2$ the minimization is performed subject to the constraint that the last $N_2 - NU + 1$ control signals are all equal. Consideration of causality of the controller implies that $\Delta u(t+NU-1)$ is the first free signal that can be calculated as a function of data available up to time $t+NU-1$ in the sequence and the argument above follows accordingly. Therefore, the GPC minimization (OLFO) is equivalent to the closed-loop policy for models of regression type.

## APPENDIX B. AN EXAMPLE FOR A FIRST ORDER PROCESS

This appendix examines the relation of $N_2$ and the closed-loop pole position for a simple example. Consider a nonminimum-phase first-order process (first-order + fractional dead-time):

$$(1 + a_1 q^{-1})y(t) = (b_0 + b_1 q^{-1})u(t-1)$$
$$(1 - 0.9q^{-1})y(t) = (1 + 2q^{-1})u(t-1).$$

### B.1. $E$ AND $F$ PARAMETERS

From the recurrence relationships in the main text it follows that:

$$e_i = 1 - a_1 e_{i-1};\ f_{i0} = 1 - f_{i1};\ f_{i1} = e_i a_1$$
$$e_0 = 1{:}[1.0];\ f_{10} = (1 - a_1){:}[1.9];\ f_{11} = a_1{:}[-0.9]$$
$$e_1 = 1 - a_1{:}[1.9];\ f_{20} = 1 - a_1(1 - a_1){:}[2.71];$$
$$f_{21} = a_1(1 - a_1){:}[-1.71]$$
$$e_2 = 1 - a_1(1 - a_1){:}[2.71];$$
$$f_{30} = 1 - a_1(1 - a_1(1 - a_1)){:}[3.439]$$
$$f_{31} = a_1(1 - a_1(1 - a_1)){:}[-2.439].$$

### B.2. CONTROLLER PARAMETERS

$$g_0 = b_0{:}[1.0];\ g_{11} = b_1{:}[2.0]$$
$$g_1 = b_0(1 - a_1) + b_1{:}[3.9];\ g_{22} = b_1(1 - a_1){:}[3.8]$$
$$g_2 = b_0(1 - a_1(1 - a_1)) + b_1(1 - a_1){:}[6.51];$$
$$g_{33} = b_1(1 - a_1(1 - a_1)){:}[5.42].$$

Assuming $NU = 1$, the effect of $N_2$ on the control calculation and the closed-loop pole position is now examined:

$$\Delta u(t) = \left(\sum g_i(w - f_{i0}y(t-1) - f_{i1}y(t-2)\right) - g_{ii}\Delta u(t-1)\sum (g_i)^2.$$

For a controller of the form $R(q^{-1})\Delta u(t) = w(t) - S(q^{-1})y(t)$

$$s_0 = \sum g_i f_{i0} / \sum g_i$$
$$s_1 = \sum g_i f_{i1} / \sum g_i$$
$$r_0 = \sum (g_i)^2 / \sum g_i$$
$$r_1 = \sum g_i g_{ii} / \sum g_i$$

and the closed-loop pole-positions are at the roots of $RA\Delta + q^{-1}SB$.

$N_2 = 1$

$$\Delta u(t) = [w - 1.9y(t) + 0.9y(t-1) - 2\Delta u(t-1)]$$
$$RA\Delta + q^{-1}SB \Rightarrow (1 + 2q^{-1}).$$

This is the minimum-prototype controller which, because of the cancellation of the nonminimum-phase zero, is unstable.

$N_2 = 2$ (*i.e. greater than deg*(B))

$$\Delta u(t) = [w - 1.9y(t) + 0.9y(t-1) - 2\Delta u(t-1) + 3.9(w - 2.71y(t) + 1.71y(t-1) - 3.8\Delta u(t-1))]/16.21$$

or:

$$\Delta u(t) = [4.9w - 12.469y(t) + 7.569y(t-1) - 16.82\Delta u(t-1)]/16.21$$

and:

$$AR\Delta + q^{-1}SB \Rightarrow (1 - 0.09q^{-1}).$$

The closed-loop pole is within the unit circle and therefore the closed-loop is stable. In fact it can easily be demonstrated that for all first order processes $N_2 > \delta B(q^{-1}) + k - 1$ stabilizes an open-loop stable plant, for $\mathrm{sign}\left(\sum g_i\right) = \mathrm{sign}(\mathrm{D.C.\ gain})$.

$N_2 = 3$

$$\Delta u(t) = [11.41w - 34.857y(t) + 23.447y(t-1) - 52.104\Delta u(t-1)]/58.591$$

and:

$$AR\Delta + q^{-1}SB \Rightarrow (1 - 0.416q^{-1}).$$

The pole is again within the unit circle. Note that the closed-loop pole is tending towards the open-loop pole as $N_2$ increases (see Part II).